

Hoa Huynh

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GitHub: github.com/huynhthaihoa

Google Scholar: scholar.google.com/citations?user=abuFJTkAAAAJ&hl=en

Seoul, South Korea

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Portfolio: huynhthaihoa.github.io

LeetCode: leetcode.com/u/huynhthaihoa

ABOUT

Computer Vision Engineer with more than seven years of experience developing and deploying real-time deep learning systems on edge devices for automotive safety and surveillance. Experienced in 2D/3D perception, inference optimization, multi-sensor integration, and edge AI deployment.

RESEARCH INTERESTS

Visual SLAM, 3D Vision, Robotics, Automotive, Edge AI Deployment

SKILLS

Programming Languages: C, C++, Python, C#, MATLAB

Perception & Deep Learning: Open3D, PCL, OpenCV, PyTorch, TensorFlow, Darknet

Edge AI Deployment: TI Edge AI/TIOVX (SK-TDA4VM, TDA4AEN), NVIDIA Jetson Nano/Xavier NX (TensorRT, DeepStream), Samsung Exynos V920, ONNX Runtime, TFLite

Sensors & Calibration: RGB/IR pinhole/fisheye, FLIR thermal, Intel RealSense, LUCID Helios2 ToF, Ouster LiDAR, ROS

MLOps & Tooling: Git, SVN, Docker, CMake, TensorBoard, DVC, W&B, MLflow, Jira, Confluence

Databases: Microsoft SQL Server, MySQL, SQLite, PostgreSQL, MongoDB

Additional Languages & Frameworks: Node.js, React, JavaScript, Java, LaTeX, WPF, Qt, FFmpeg

EXPERIENCE

DeltaX

Seoul, South Korea

Development Team Lead & AI Researcher

June 2022 – July 2026

- Led an R&D team from December 2023, driving the architecture, technical roadmap, and end-to-end delivery of an In-Cabin Monitoring System (ICMS) that aligned with Euro NCAP standards, while coordinating module development, milestone tracking, and client demonstrations across cross-functional teams.
- Architected a real-time edge AI pipeline using a single RGB/IR fisheye camera, C/C++ inference on an SoC (System-on-Chip), and a Node.js/React web visualization UI.
- Integrated 8 deep learning models (object detection, 2D/3D body keypoint estimation, facial landmark detection, gaze estimation, and image classification), achieving an average of 14 FPS on an 8-TOPS edge platform and 9 FPS on a 4-TOPS edge platform.
- Delivered 4 Proof-of-Concept (PoC) projects for 2 Korean automotive vendors within 3–9-month timelines: 42dot (Phase 1: January – June 2024; Phase 2: August – October 2024), Pyeong Hwa Automotive (Phase 1: May 2024 – January 2025; Phase 2: July – October 2025).
- Customized PyTorch models and optimized Python/C++ integration pipelines for SoC targets, deploying on TI SK-TDA4VM/TDA4AEN using Edge AI + TIOVX frameworks, and on Samsung Exynos V920 using an ONNX Runtime-based framework.
- Developed the 3D perception stack for worker localization, 3D display, and ADAS applications (Advanced Driver-Assistance Systems): monocular depth estimation (with pseudo-LiDAR conversion) trained under LiDAR/RGB-D supervision, semantic segmentation, visual SLAM, and real-time multi-camera stitching from fisheye and pinhole inputs.
- Designed a slicing-based preprocessing scheme derived from SAHI (“Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection”) to raise recall on small objects for YOLOv8-p2/YOLOv9n models while maintaining real-time inference latency.
- Built the team’s multi-sensor calibration, data collection, and analysis tooling (ROS, MATLAB, Python), and assembled real-world ADAS/in-cabin datasets across diverse subjects and lighting conditions.

Smartdeep

Research Engineer

Seoul, South Korea

May 2021 – May 2022

- Built a 16-channel CCTV surveillance system in C++ (OpenCV, FFmpeg) with a C# WPF front end, aligned with KISA Intelligent CCTV certification requirements for moving- and abnormal-object detection, recording, and event search.
- Owned dataset curation, training, and validation pipelines for Darknet YOLOv4 detection models.

Ubay Solution

AI Research Intern

Seoul, South Korea

December 2020 – April 2021

- Developed an optical character recognition (OCR) system for deployment at Busan Port, enabling automated detection and recognition of truck license plate numbers and container codes.
- Benchmarked object detection models (SSD, YOLOv4, Faster R-CNN, etc.) on a custom dataset.
- Applied data augmentation and web crawling techniques to increase dataset size and quality.
- Implemented a DeepStream + TensorRT pipeline on NVIDIA Jetson Nano and Jetson Xavier NX for model deployment and real-time inference.

ANDA Lab – Soongsil University

Research Assistant

Seoul, South Korea

September 2018 – August 2020

- Conducted research and experimental analysis using Python, OpenCV, and MATLAB to diagnose issues in a vehicle localization system centered on taillight detection and processing.
- Co-authored papers proposing solutions to address the identified issues.

Global Cybersoft JSC

Software Engineer Trainee

Ho Chi Minh City, Vietnam

March 2017 – August 2017

- Developed C++ Microsoft Foundation Classes (MFC) plugins for SCREEN Holdings Co.'s industrial printer system (Japan).
- Developed Adobe Illustrator plugins using the Adobe Illustrator SDK, C++, and JavaScript for an advertising agency (Japan).
- Improved code quality and team collaboration by participating in Jira-based peer code reviews.
- Gained hands-on experience in an Agile software development environment.

EDUCATION

Soongsil University

Master of Engineering, Information and Communication Technology

Seoul, South Korea

September 2018 – August 2020

- **GPA:** 4.01/4.5
- **Thesis:** LED Taillight Matching Algorithm for Nighttime Vehicle Positioning
- **Ref:** oasis.ssu.ac.kr/search/i-discovery/edsker:edsker.000004851619?type=eds (English)
- **Supervisor:** Professor Myungsik Yoo

University of Information Technology – Vietnam National University

Bachelor of Engineering, Software Engineering

Ho Chi Minh City, Vietnam

August 2013 – June 2018

- **GPA:** 8.6/10 (Salutatorian Honors)
- **Thesis:** Blood Vessel Segmentation on DICOM Images
- **Ref:** drive.google.com/file/d/1RylFqRy-xywZG1zEWEX2l3B49T7XbFvf/view (Vietnamese)
- **Supervisor:** Professor Thi-Vuong Pham

PUBLICATIONS

(**J**: International Journal, **W**: International Workshop, **K**: Korean Journal/Conference)

K. Uddin S M Nadim et al. (including **Huynh Thai Hoa**), “A Railway Framework for Person (Worker) Localization based on Depth Estimation and GPS”, *The Korean Institute of Defense Technology Fall Conference* (2024). Ref: kidet.or.kr/index.php?page=view&pg=&idx=1259&hCode=BOARD&bo_idx=1&sfl=&stx=

W. Jamie Spencer et al. (including **Huynh Thai Hoa**), “The Second Monocular Depth Estimation Challenge”, *Conference on Computer Vision and Pattern Recognition Workshops* (2023). Ref: arxiv.org/abs/2304.07051

J. Thai-Hoa Huynh, Myungsik Yoo, “A Taillight Matching and Pairing Algorithm for Stereo-Vision-Based Night-time Vehicle-to-Vehicle Positioning”, *MDPI Applied Sciences* (2020). Ref: mdpi.com/2076-3417/10/19/6800

K. Thai-Hoa Huynh, Myungsik Yoo, “Nighttime LED Taillight Detection Method with Optical Camera Communication System”, *The Journal of Korean Institute of Communications and Information Sciences* (2020). Ref: dbpia.co.kr/journal/articleDetail?nodeId=NODE10440000

J. Thai-Hoa Huynh, Tuan-Anh Pham, Myungsik Yoo, “Detection Algorithm for Overlapping LEDs in Vehicular Visible Light Communication System”, *IEEE Access* (2019). Ref: ieeexplore.ieee.org/document/8792184

SELECTED PROJECTS

State Estimation Simulations: from-scratch Lie-group state-estimation simulations – EKF, Invariant EKF, UKF, factor-graph and incremental (iSAM) optimization, bundle adjustment. [GitHub](#)

Monocular Metric Depth Estimation: PyTorch pipeline estimating per-pixel metric depth from a single image, with pluggable encoders (EfficientNet, ConvNeXt, DINOv2, YOLO-based) and decoders (NeWCRFs, DPT, PixelFormer), trained on NYUv2/KITTI dataset with EdgeAI-TIDL-based quantization-aware training support and deployed via ONNX/TFLite export. [GitHub](#)

Notch-Based Point Cloud Registration: C++ point-cloud registration aligning 3D scans to a reference model via notch-keypoint correspondence matching, for industrial inspection and quality verification. [GitHub](#)

LiDAR-Based Ego Lane Detection: classical signal-processing pipeline (ground extraction, intensity analysis, polynomial fitting) for interpretable ego-lane detection directly from LiDAR point clouds. [GitHub](#)

Vision Encoder Playground: curated collection of pre-trained vision encoder (backbones) from detection and classification models, exposed as reusable feature extractors for downstream computer vision tasks. [GitHub](#)

(Full project list available on portfolio: huynhthaihoa.github.io/#projects)

SELECTED CERTIFICATIONS

Hugging Face:

- Fine-tuning Language Models (2025)
- AI Agents Fundamentals (2025)
- Fundamentals of Group Relative Policy Optimization (2025)

Coursera:

- Introduction to Machine Learning in Production (2024)

HackerRank:

- Problem Solving (Intermediate) (2024)

LeetCode:

- 500 Days Badge (2026)
- Introduction to Pandas (2024)

Kaggle:

- Computer Vision (2022)
- Advanced SQL (2022)
- Intermediate Machine Learning (2020)

(Full certification list available on portfolio: huynhthaihoa.github.io/#certifications)